

EXPLORATORY DATA ANALYSIS (EDA) REPORT

Project : Advanced Regression Model Comparison & Optimization Platform

Dataset : House Prices - Advanced Regression Techniques

1. Business Objective

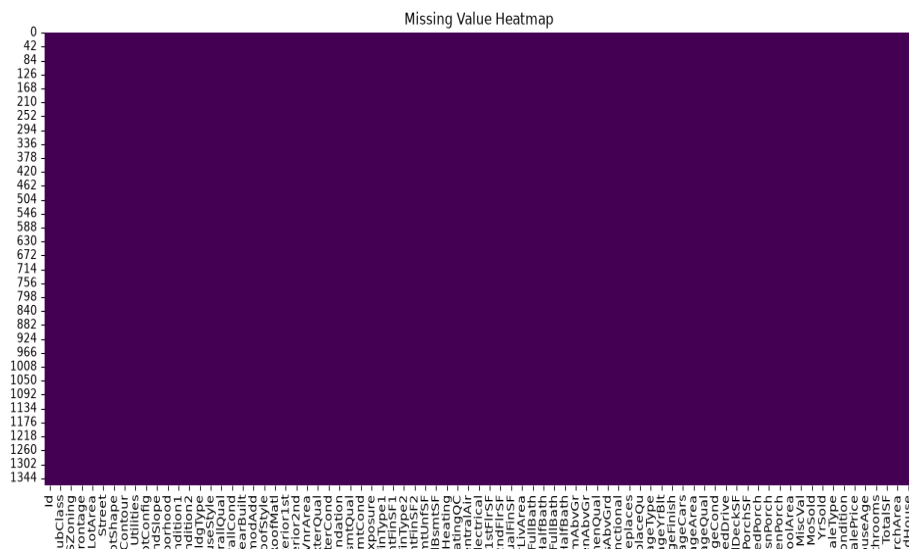
The objective of this project is to analyze housing data, identify the factors affecting house prices and understand the relationships between different property features before building regression models. Exploratory Data Analysis helps discover data quality issues, feature distributions, missing values and correlations that influence prediction accuracy.

2. Dataset Summary

Attribute	Value
Dataset	House Prices
Total Records	1460
Total Features	80
Target Variable	SalePrice
Numerical Features	37
Categorical Features	43

3. Missing Value Analysis

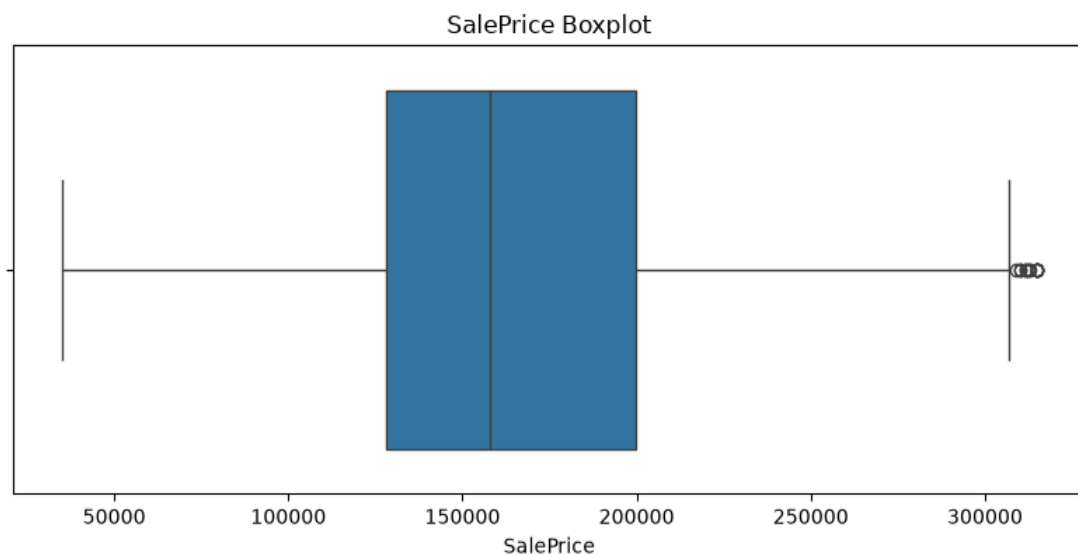
The dataset contained missing values in several numerical and categorical columns. Before model development, these missing values were carefully analyzed and later handled using appropriate preprocessing techniques.



Observation: Missing values were mainly concentrated in a few features, indicating the need for data imputation before model training.

4. Target Variable Distribution

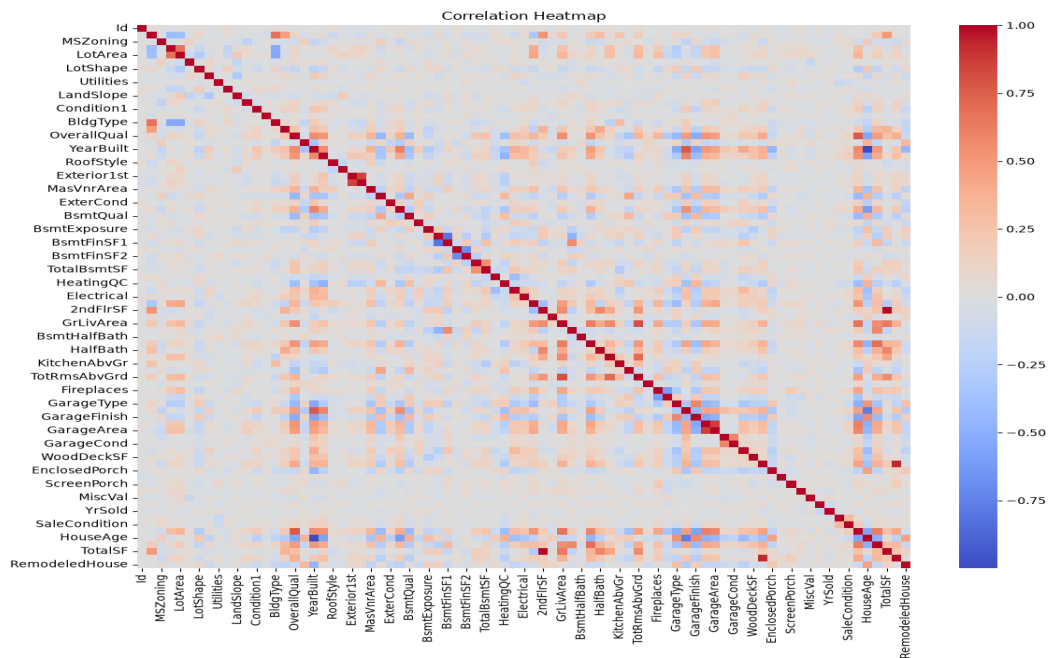
Understanding the target variable distribution is important before building regression models. The SalePrice distribution helps identify skewness and the presence of extreme values.



Observation: SalePrice shows a positively skewed distribution, suggesting that transformation techniques such as log transformation may improve model performance.

5. Correlation Analysis

Correlation analysis identifies relationships between numerical variables and the target variable. Highly correlated features generally contribute more to prediction accuracy.



Observation: OverallQual, GrLivArea, GarageCars and TotalBsmtSF show strong positive correlation with SalePrice and are important predictive features.

6. Key Findings

- The dataset contains both numerical and categorical features.
- Missing values were identified and prepared for preprocessing.
- The target variable exhibits positive skewness.
- Several housing characteristics have a strong influence on sale price.
- Exploratory analysis provided valuable insights for feature selection and model development.

7. Conclusion

Exploratory Data Analysis provided a comprehensive understanding of the dataset by identifying missing values, understanding feature distributions and discovering relationships between important variables. These insights guided the preprocessing stage and contributed to building more accurate regression models for house price prediction.